

Polytropes and the Lane-Emden Equation

Daniel Hilfeldt, Adam Quayle

Department of Physics, Maynooth University

April 2026

ABSTRACT

The Lane-Emden equation was solved numerically over the real numbers for polytropic indices corresponding to physical celestial bodies, as well as over the complex numbers for those indices which allow exact solutions. The physical implications of these solutions on solar structure were explored via the Eddington standard model's assumption of a solar polytropic index of $n = 3$. This exploration included the calculation of solar density, mass, temperature, pressure and luminosity, which were then compared to the standard solar model's estimates of these parameters. Also included is a discussion of the computational methods used, such as Euler, Euler-Cromer and 4th-order Runge-Kutta, and an examination of non-physical solutions to the Lane-Emden equation.

1 Introduction

STELLAR structure is dictated by a set of fundamental differential equations which can be solved to gain powerful insight into the properties of stars. The solutions to these fundamental equations, however, require one parameter to already be known — the density function $\rho(r)$.

Mathematically, the density function can take the form of any integrable function, although not every integrable function is guaranteed to produce physical solutions by any means. One set of solutions which does approximate physical reality is derived from the assumption that the stellar material is polytropic. Under this assumption, and a few other more minor assumptions, the equations of stellar structure reduce to the *Lane-Emden equation*, the solutions to which provide an alternative route to obtaining the elusive stellar density function.

The aim of this study is to find solutions to the Lane-Emden equation, also known as *polytropes*, computationally. In the process, an array of ODE solvers will be compared to each other and tested

against the known analytical solutions to the Lane-Emden equation. The polytropes obtained can then be used to predict known stellar parameters.

2 Prerequisite Theory

2.1 Polytropic Processes

A polytropic process is one in which the specific heat of the medium remains constant [1]. If this is the case, then the relationship in Equation 1 holds true¹.

$$PV^\beta = C \quad (1)$$

If the process which keeps the specific heat constant is the movement of material from one place to another, then the equation of state detailed in Equation 1 describes a constant of the material which holds across the entire medium. With some simple manipulation, and the substitution $\beta \rightarrow 1 + 1/n$, Equation 1 can be written as

$$P = K\rho^{1+\frac{1}{n}} \quad (2)$$

The *polytropic index* n is a constant which depends on the medium being considered. If the medium can be approximated by an ideal gas, then many well-known thermodynamic processes are actually members of the polytropic family — such as isobaric processes ($\beta = 0$), isothermal processes ($\beta = 1$) and adiabatic processes ($\beta = \gamma$)

Of particular note to this paper is the fact that stars can be approximately described by the polytropic equation of state (Equation 2). Different polytropic indices can be used to model the interior of various celestial bodies

2.2 The Lane-Emden Equation

If the assumption of a polytropic equation of state is applied to the equations of stellar structure, then they can be reduced to the *Lane-Emden Equation*.

$$\frac{1}{\xi^2} \frac{d}{d\xi} \left(\xi^2 \frac{d\theta}{d\xi} \right) + \theta^n = 0 \quad (3)$$

(For the derivation, see Appendix B1).

¹A glossary of symbols is available in Appendix A

Each solution of the Lane-Emden equation (for a given n) is called a *polytrope*.

In a coordinate system with its origin at the barycentre of the star, ξ is a dimensionless measure of the radial coordinate r and θ is a dimensionless measure of the density ρ . Both are defined with respect to a characteristic scale of their respective quantities.

$$r = \alpha \xi = \xi \cdot \sqrt{\frac{(n+1)K\rho_c^{\frac{1}{n}-1}}{4\pi G}} \quad (4)$$

$$\rho = \rho_c \theta^n \quad (5)$$

Apart from the special cases of $n = 0$ and $n = 1$, Equation 3 is a non-linear second-order ordinary differential equation.

There are only three polytropic indices for which exact solutions exist for the Lane-Emden equation. Unsurprisingly, two of these are the indices which cause the equation to become linear, $n = 0$ and $n = 1$. Interestingly, $n = 5$ also produces an exact solution [2].

Numerical methods are required to find solutions for all other polytropic indices. The

goal of this paper is to explore a variety of these methods and use the computed solutions to produce a model of the stellar interior.

3 Computational Modelling

3.1 Lane-Emden Solutions

The Lane-Emden equation is given, in its standard form, as

$$\frac{d^2\theta}{d\xi^2} + \frac{2}{\xi} \frac{d\theta}{d\xi} + \theta^n = 0 \quad (6)$$

with a singular point at $\xi = 0$. This equation has analytic solutions for $n = 0, 1$, and 5 (see Appendix B2). However, outside of these, the solution must be estimated either by series expansion or, as done here, numerically [3].

A range of numerical ODE solvers can be used to compute the solution. Four such methods, the Euler, Euler-Cromer, Midpoint, and 4th order Runge-Kutta, are shown in Figure 1. For these computations, the initial conditions of $\theta(0) = 1$ and $\frac{d\theta}{d\xi}(0) = 0$, were given, corresponding to a constant central density of the star.

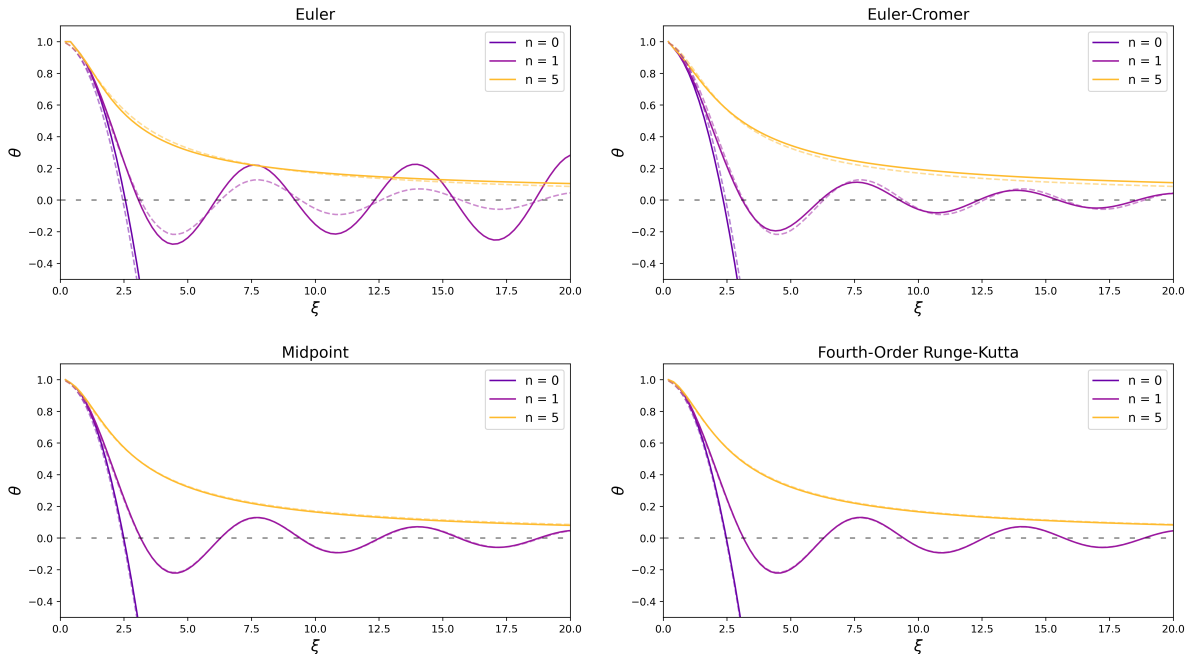


Figure 1: Different numerical solutions to the Lane-Emden equation for polytropic indices $n \in \{0, 1, 5\}$ (solid), along with the corresponding analytical solutions (dashed). In all cases, the step size used is 0.2.

A relatively large step size was used to highlight the differences between the methods. The Euler method is notably inaccurate at dealing with oscillating function, such as the $n = 1$ case. This is largely fixed with the Euler-Cromer method, and a greater accuracy overall is achieved by the Midpoint and furthermore 4th-order Runge-Kutta methods. This improved accuracy is, however, at the cost of an increased number of computations needed for each step. Since the function evaluation of the Lane-Emden equation is not computationally heavy, the Runge-Kutta 4th Order method

is used to compute the numerical solutions for $n \in \{0, 1, 2, 3, 4, 5\}$, as shown in Figure 2.

The solutions of the Lane-Emden equation are seen to satisfy the initial conditions of $\theta(0) = 1$, $\frac{d\theta}{d\xi}(0) = 0$, followed by a rapid decline. The cases of $n = 0, 2$ continue to decline, while the solutions for $n = 3, 4, 5$ express a stabilisation towards $\theta = 0$. Notably, the case of $n = 1$ shows an oscillatory behaviour around $\theta = 0$. For further comments on the asymptotic behaviour of the Lane-Emden solutions, see subsection 4.2.

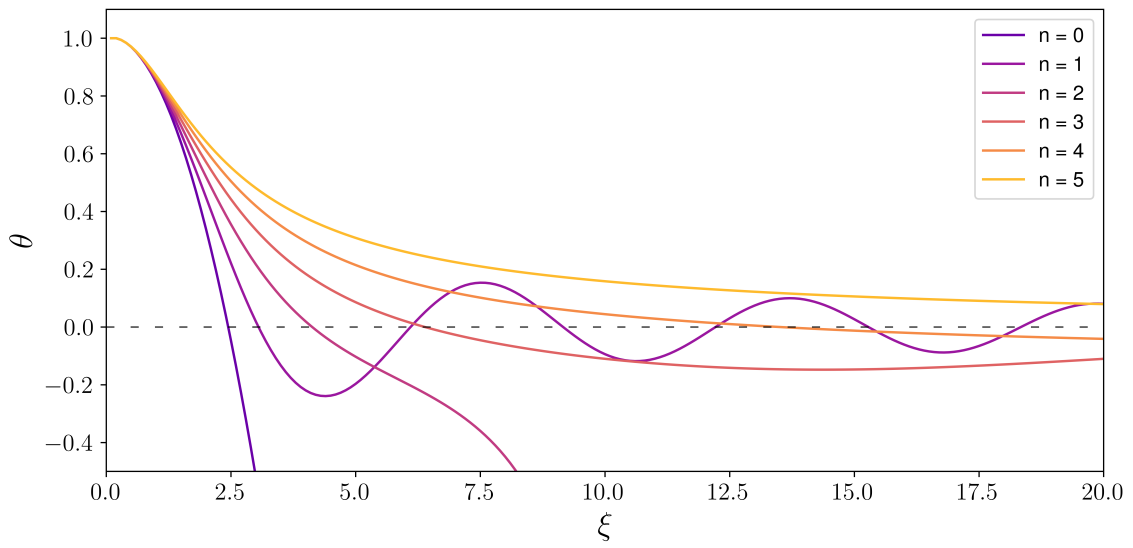


Figure 2: Solutions to the Lane-Emden equation for integer values of n .

3.2 Fractional Polytropic Indices

The solutions shown in Figure 2 are of integer polytropic index. However, if the differential equation is considered for a fractional, non-integer polytropic index, $n = \frac{a}{b} \in \mathbb{Q} \setminus \mathbb{Z}$, Equation 6 can be rewritten as

$$\frac{d^2\theta}{d\xi^2} + \frac{2}{\xi} \frac{d\theta}{d\xi} + (\sqrt[b]{\theta})^a = 0 \quad (7)$$

As such, for negative values of θ , Equation 6 becomes a complex differential equation.

Since the ODE solvers used work solely in a real domain, the evaluation of the solution at negative values of θ results on an evaluation error, and a truncation of the integration process, as seen in Figure 3.

However, as this study focuses on physical solu-

tions of the Lane-Emden equation, and a negative density is non-physical, the solution should not be considered relevant beyond this point if only stellar structure is being considered. Specifically, the point at which $\theta \rightarrow 0$ should be interpreted as the edge of the star. (see subsection 4.2 for details)

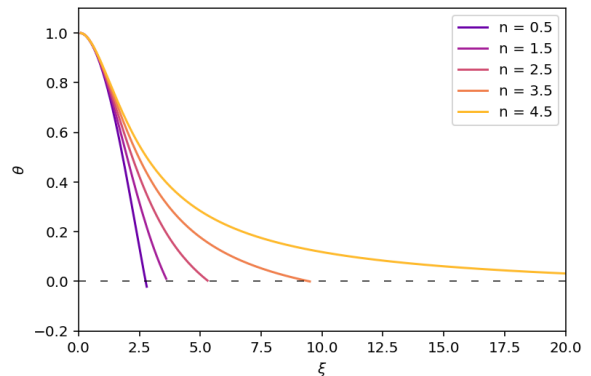


Figure 3: Fractional Polytrope Solutions to the LE-Equation

3.3 Complex Solutions on a Real Domain

Looking past the non-physicality of a negative density, the ODE solver used can easily be adapted to include complex solutions to the Lane-Emden equation. As such, solutions of the form $\theta : \mathbb{R} \rightarrow \mathbb{C}$ can be computed. That is, the polytrope solution has a real domain, and a complex co-domain. The set of solutions from Figure 3 is shown in Figure 4, with the imaginary and real parts plotted.

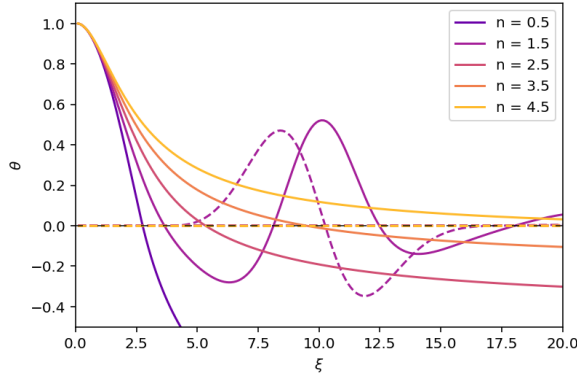


Figure 4: Real (solid) and Imaginary (dashed) parts of the complex solutions to the Lane-Emden equation

The $n = \frac{3}{2}$ case behaves as expected, with an imaginary contribution as the real part of the dimensionless density crosses the $\theta = 0$ threshold. However, the imaginary parts of the other 4 solutions seem negligible. Further commentary on this avenue of research is given in Appendix C.

4 The Solar Interior

4.1 A Note on Units

For the sake of computational accuracy and generality, the quantities in this paper will be reported in units of solar masses $M_\odot$, solar radii $R_\odot$ and solar luminosities $L_\odot$. This allows computation to be carried out without introducing the potential inaccuracy brought by consulting physical measurements. In some places, if useful, other units may be used — but a note will be made if this is the case. The standard units for both temperature and pressure in this paper are $T_{c,\odot}$ and $P_{c,\odot}$, the temperature and pressure in the core of the Sun respectively.

An example of a derived unit in this system is stellar density ρ . The average density of the Sun is definitionally $\rho_{\text{av}} = M_\odot/V_\odot = 3M_\odot/4\pi R_\odot^3$. So, in the units described above, $\rho_{\text{av}} = \frac{3}{4\pi}$. This value is used to annotate the mean solar density in Figure 6.

4.2 Roots of Solutions

The solutions found to the Lane-Emden equation in subsection 3.1 are subject to some simple physical restrictions which heavily reduce the domain of the density functions.

Firstly, the density must be non-negative at every point in the star, only the surface of a star can be approximated as being at a vacuum pressure. Furthermore, once the density falls to zero, it must remain at zero, there cannot be any regions of vacuum within a star in hydrostatic equilibrium, as the stellar pressure would collapse the cavity within a time smaller than the free-fall timescale.

These two rules imply that the domain of the solutions to the Lane-Emden equation should be restricted to $[0, \xi_R]$ where ξ_R is the first root of the function. These roots can be found by starting at $\xi = 0$ and recording the value of ξ where θ goes from being positive to negative in Figure 2. This data is displayed in Table 1, and agrees with values quoted in the literature [4].

The position of the first root of the density function defines the boundary of the star, allowing the value of α in Equation 4 to be set. One stellar radius corresponds to the first root of the density function for a given n (see Table 1). So Equation 4 becomes

$$\alpha = \frac{R}{\xi_R} \quad (8)$$

This definition of α becomes particularly clean when the radial co-ordinate r is given in units of stellar radii, as then α is simply the reciprocal of ξ_R .

n	0	1	2	3	4	5
ξ_R	2.45	3.14	4.35	6.89	14.96	—
ξ_R^*	$\sqrt{6}$	π	—	—	—	∞

Table 1: Roots ξ_R of the function θ for a polytropic index n , and the associated exact roots ξ_R^* if an analytical solution to the Lane-Emden equation exists.

It can be seen in Table 1 as well as in Figure 2 that the first root of the density function ξ_R seems to increase with the polytropic index n . This increase is, in fact, monotonic, and $n = 5$ is the final index for which the density function asymptotes to zero [5]. Furthermore, as a function must at least asymptote to zero for its improper integral to be convergent,

$n = 5$ is the final case of finite stellar mass. So, a polytropic index of $n = 5$ corresponds to a star with infinite radius but finite mass, $n < 5$ corresponds to a star which is finite in both radius and mass, and $n > 5$ corresponds to a star which is infinite in both.

4.3 The Standard Solar Model

The standard solar model (SSM) is likely the most sophisticated model of the Sun's interior which is currently available. Unlike the polytrope model outlined in this paper, the SSM takes account of the nuclear fusion and time-varying mass fractions of the solar interior. It is able to accomplish this by beginning with the Sun as a $1M_{\odot}$ protostar and evolving it through time via the equations of stellar structure, stopping at the current age of the Sun.

The equations of stellar structure are a set of coupled first-order differential equations, meaning that they require some information to be provided in the form of boundary conditions to set the scale of the solutions, this data is obtained through physical observation of the Sun. One advantage to this method of simulation is that the solar parameters of the past and future can also be accurately determined by stopping the evolution earlier or later.

The way in which energy is transported around the interior is very important for both the structure and evolution of the star. The two methods by which this is achieved are convection and radiation, and the type of transport which dominates depends on the internal pressure and temperature functions (as well as some stellar constants). The SSM is able to predict accurately the radius at which the Sun changes from being radiative to convective using the predicted pressure and temperature within the Sun. The polytrope model on the other hand predicts that stars of $n > 1.5$ are totally radiative and stars of $n < 1.5$ are entirely convective (see Appendix B3). While there do exist stars like this, it's not true that *all* stars are dominated by only a single method of energy transport — our Sun is one of many counter-examples.

In the context of this paper, the SSM can act as a quality reference to which the accuracy of the polytrope approximation can be compared. Due to this resource being available, most of the polytrope models in this section will be

of $n = 3$, which the Eddington solar model uses as a close approximation for the Sun [6].

4.4 The Enclosed Mass Function

After solving the Lane-Emden equation for a given polytropic index (subsection 3.1), the density at a given radius r can be obtained via the definition of θ (Equation 5). The enclosed mass of the star at a given radius can then be obtained by integrating this density over the volume contained within a sphere of radius R . As stars are spherically symmetric, only the radial component need be involved in the integration (which simplifies the computation). The differential equation which describes the mass enclosed by a sphere of radius r is

$$\frac{dM_r}{dr} = 4\pi r^2 \rho \quad (9)$$

The relevant integral is then

$$M_R = 4\pi \int_0^R r^2 \rho(r) dr \quad (10)$$

There are many ways to tackle this integral numerically, but for the sake of simplicity and variety, the enclosed mass displayed in Figure 5 has been calculated via the Euler method — the conceptually simplest integration technique.

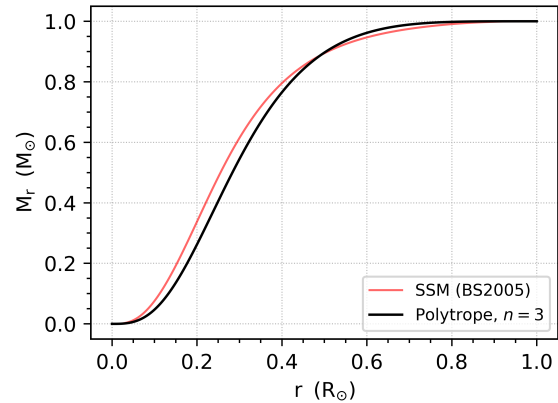


Figure 5: The enclosed mass function $M(r)$ of the Sun. The polytrope model ($n = 3$) is compared against the standard solar model. [7][8]

4.5 Solar Density

As mass is an extensive property, it brings along with it a normalisation condition for the enclosed mass function $M_r(r)$. The enclosed mass at the surface of the star must be the total mass of the star. In this case, as the mass of the Sun is taken as the standard unit of mass, the density function must be normalised such that the integral over the volume of the Sun equals 1.

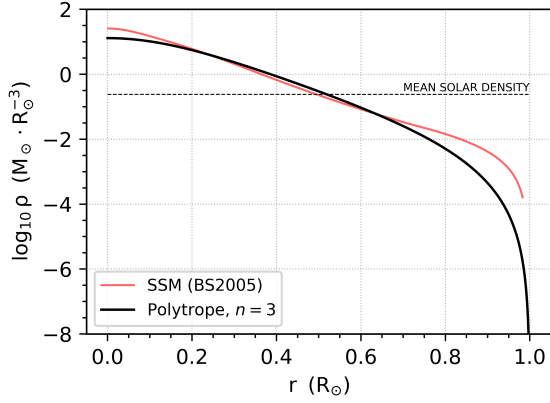


Figure 6: The predicted density function of the Sun

4.6 Solar Pressure and Temperature

The pressure and temperature at a given radius within the star are directly calculable from the density function $\rho(r)$. The pressure function $P(r)$ for a given polytropic index can be found by recalling the polytropic equation of state (Equation 2):

$$P \propto \rho^{1+\frac{1}{n}}$$

The internal temperature function of the star is derived from the ideal gas law.

$$P = \rho \frac{k_B T}{\mu m_H} \quad (11)$$

$$T = \frac{\mu m_H P}{k_B \rho} \quad (12)$$

Then, Equation 2 can be used to replace P

$$T = \frac{\mu m_H}{k_B} \rho^{\frac{1}{n}} \quad (13)$$

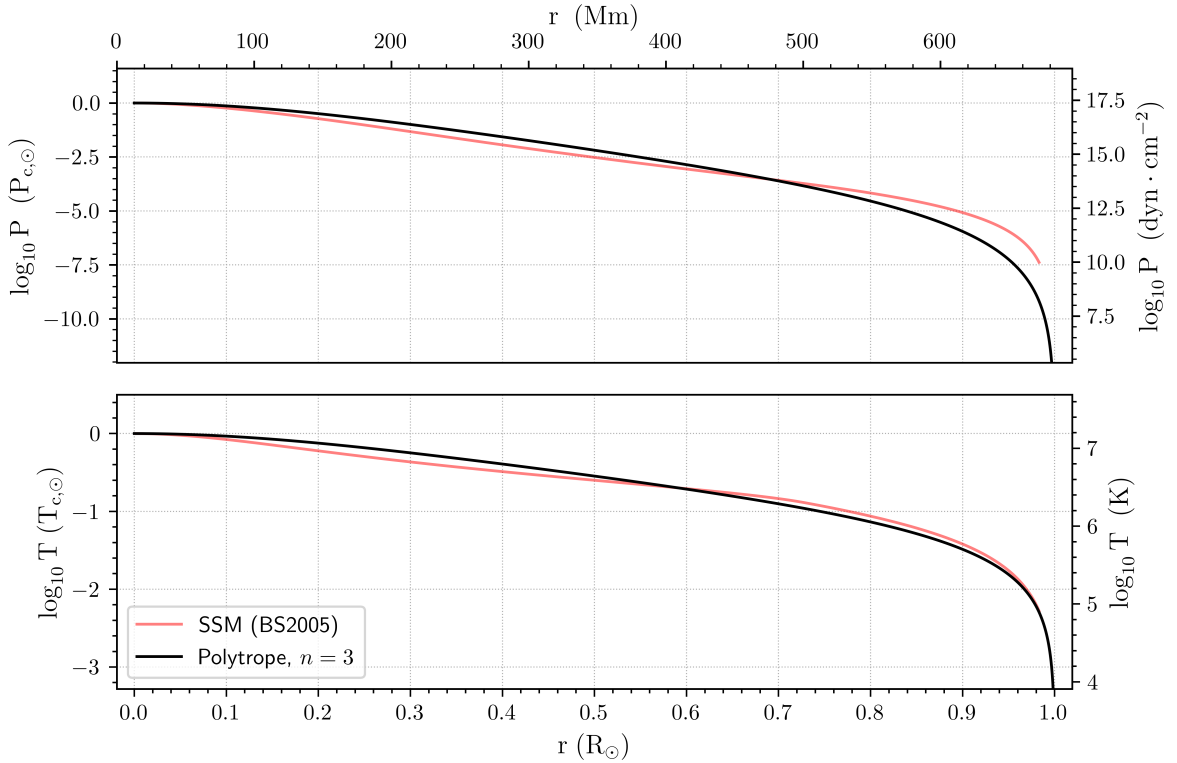


Figure 7: TOP: The internal pressure of the Sun given by both the SSM and the polytrope model. BOTTOM: The temperature of solar material predicted by the same models. The SSM database [8] quotes quantities in cgs units, which are also displayed.

4.7 Solar Luminosity

The luminosity of stellar material is described by a fundamental equation of stellar structure.

$$\frac{dL}{dr} = 4\pi r^2 \rho \epsilon \quad (14)$$

Evidently, this is similar to the equation describing mass conservation (Equation 9). This makes sense, as ϵ describes the amount of power emitted *per mass* within the star. This equation would be handled trivially if ϵ were a constant across the star. Energy generation is, however, not homogeneous throughout stellar interiors (although the exact distribution and type of fusion depends largely on the mass of the star). Within the Sun, only the core becomes hot enough to fuse hydrogen into helium via the proton-proton chain [9].

One crude approximation of the $\epsilon(r)$ function is a step function which transitions at the temperature where the proton-proton chain begins to dominate. To be more specific, most of the energy of the Sun is generated via the p-pI chain which begins to act at around ~ 10 MK [10]. While the other branches of the proton-proton chain are active within the Sun, the magnitude of error introduced via the step function approximation will likely dominate any perturbation in the profile caused by the action of non-primary fusion processes.

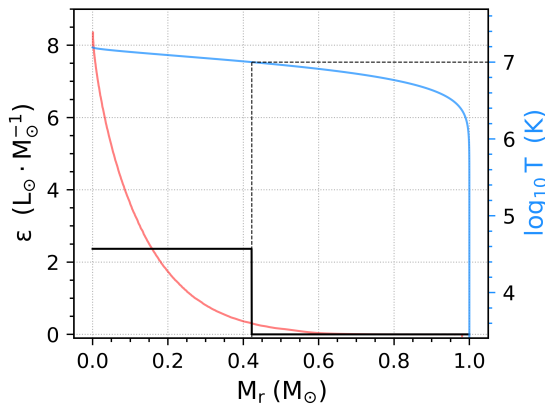


Figure 8: BLACK: Approximated power per unit mass from the polytropic model. RED: Power per unit mass from the standard solar model (subsection 4.3). BLUE: Polytropic approximation of stellar temperature (subsection 4.6), 10MK being the point at which the p-pI chain is estimated to begin.

The approximated specific power function ϵ displayed in Figure 8 can then be integrated to

find the luminosity of the solar material enclosed by a sphere of radius r . Expressing the x -axis in units of enclosed mass will prove computationally useful, as all constants in Equation 14 will reduce to unity — as will be shown.

Recall

$$\frac{dM_r}{dr} = 4\pi r^2 \rho \quad (8)$$

This can then be applied to Equation 14.

$$\frac{dL}{dr} = \epsilon \frac{dM_r}{dr} \quad (15)$$

$$dL = \epsilon dM_r \quad (16)$$

$$L = \int_0^{M_r} \epsilon dM'_r \quad (17)$$

So, the luminosity function can be found through direct integration if an argument of enclosed mass is allowed. Fortunately, the solar enclosed mass function has already been estimated for a given radial distance (subsection 4.4), and so the solar luminosity as a function of the radial coordinate r can be graphed by plotting against the array of radial values corresponding to each enclosed mass value. This is done in Figure 9.

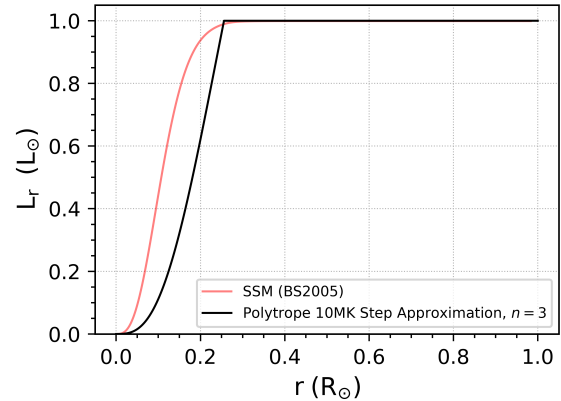


Figure 9: The luminosity enclosed within a sphere of radius r according to the 10MK step function approximation previously described, and the actual luminosity profile of the Sun according to the SSM.

Given that the polytropic estimate does not have any information about stellar fusion, the estimated luminosity profile being at least *comparable* with the SSM profile is somewhat impressive. Of course, information about the p-pI chain is encoded in the 10MK transition of the step function, but the point at which the Sun achieves that temperature is obtained only from the polytropic model.

5 Further Analysis

While finding solutions on the complex plane to the Lane-Emden equation for an arbitrary polytropic index is beyond the scope of this study, the exact solutions with physical constraints imposed ($n \in \{0, 1, 5\}$) can be analysed in the complex plane with relative ease.

In this section, the solutions $\theta : \mathbb{C} \rightarrow \mathbb{C}$ have a two dimensional domain *and* a two dimensional co-domain. This introduces the problem of displaying four dimensions of information in a Universe which only has three spatial dimensions. There are a few different solutions to this problem, including displaying the domain and codomain in two separate plots. For the sake of graphical compactness, the phase of the solution has been encoded in the colour of the plot.

In the case of $n = 0$ the Lane-Emden equation reduces to a linear ODE with an exact solution of

$$\theta = 1 - \frac{\xi^2}{6}$$

(See Appendix B2 for the derivation)

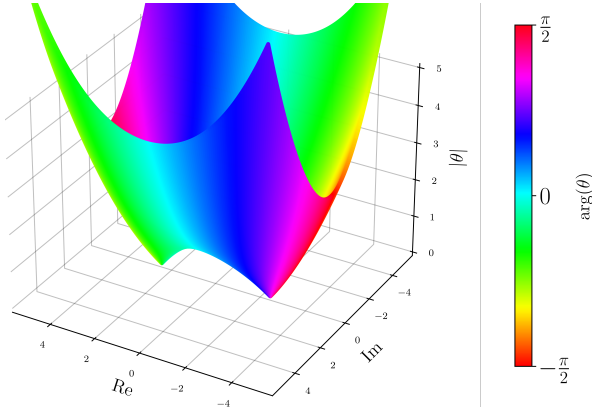


Figure 10: The exact complex $n = 0$ solution.

The solution displays the quadratic behaviour expected from the function. By the form of the exact solution, it can be seen that this function has no poles and no removable discontinuities — it is holomorphic. For this reason, $n = 0$ could be described as the most “well behaved” exact solution out of the three.

The $n = 1$ case corresponds to the high-frequency oscillating solution in Figure 2. The exact solution to the Lane-Emden equation for this polytropic

index happens to be the cardinal sine function.

$$\theta = \frac{\sin \xi}{\xi} = \text{sinc}(\xi)$$

(See Appendix B2 for the derivation)

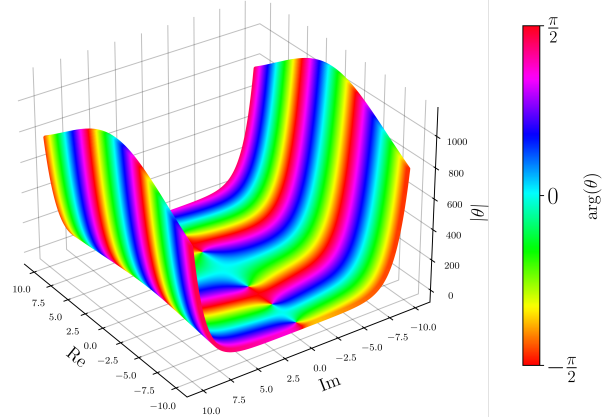


Figure 11: The exact complex $n = 1$ solution.

Technically, the function $\frac{\sin(\xi)}{\xi}$ is not defined at $\xi = 0$. This discontinuity is removable, however, and so while the function is not holomorphic, it is holomorphically extendable over the origin. As the function doesn’t blow up to infinity at the origin, this point is not considered a pole and so is not of much note. For the purposes of simplicity, the sinc function is often defined as the holomorphic extension of $\frac{\sin \xi}{\xi}$, giving the following power series definition [11].

$$\text{sinc}(\xi) = \sum_{n=0}^{\infty} \frac{(-1)^n \xi^{2n}}{(2\xi + 1)!} \quad (18)$$

The oscillations of the sinc function are what cause the periodic phase change across the real axis of the plot in Figure 11. As the z -axis represents the magnitude of the complex image, the z -value can never be negative, rather the negative parts of the sinc function are represented by colours which correspond to negative phase values.

If the value of ξ is ever purely imaginary, then the sine function in the solution can be written as a hyperbolic sine function instead (which is always real). This explains the cyan “ribs” travelling parallel to the imaginary axis in Figure 11 — the input to the sine function becomes purely imaginary at every multiple of 2π due to its periodicity.

The largest polytropic index for which there exists an exact solution is $n = 5$. In many ways, this solution is unlike the other two exact solutions. Firstly, it's the only solution for which the Lane-Emden equation is non-linear and secondly, its existence is not immediately obvious — taking almost forty years to be discovered after the Lane-Emden equation was originally proposed [12][2].

There exists two singularities in the domain of the solution which limit the radius of convergence of a power series around the origin, and restricts the open set upon which the function is holomorphic. Unlike the other two exact complex solutions, which are holomorphic everywhere, the solution for $n = 5$ can only be described as *meromorphic* due to the existence of these isolated poles. Interestingly, the position of these poles along the imaginary axis can be changed by varying the initial conditions of the IVP [13]. This exact solution, as given by Emden [2], is

$$\theta = \frac{1}{\sqrt{1 + \frac{\xi^2}{3}}}$$

(See Appendix B2 for the solution's verification.)

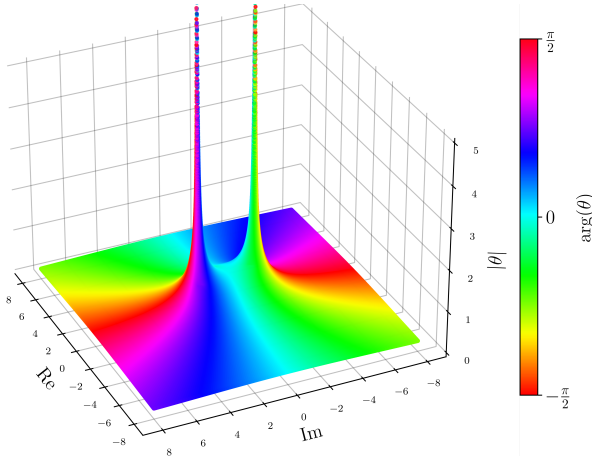


Figure 12: The exact complex $n = 5$ solution. Note the two singularities along the imaginary axis.

In the case of the initial conditions which correspond to the exact solution detailed above, *i.e.* the physical initial conditions $\theta(0) = 1$, $\left. \frac{d\theta}{d\xi} \right|_{\xi=0} = 0$, the position of the singularities can be easily found.

$$\theta = \frac{1}{\sqrt{1 + \frac{\xi^2}{3}}} \implies 1 + \frac{\xi^2}{3} = 0 \text{ at the poles.}$$

²The index $n = 3/2$ is a non-relativistic approximation. If relativity is considered, $n = 3$ is likely more accurate [14].

$$\xi_{\text{pole}}^2 = -3 \quad (19)$$

$$\xi_{\text{pole}} = \pm\sqrt{-3} \quad (20)$$

$$\xi_{\text{pole}} = \pm i\sqrt{3} \quad (21)$$

6 Remaining Avenues of Research

While the polytropic index $n = 3$ was of particular focus in this paper, this index isn't unique in corresponding to a approximation of a physical type of star. Below is a table of polytropic indices and the type of body which they approximate the behaviour of.

n	BODY
0	Rocky/Liquid Planets (Constant Density)
1.5 ²	White Dwarfs [14]
3	Main Sequence Stars [6]
∞	Globular Clusters (Isothermal)

Table 2: The polytropic indices which best model some celestial objects or systems.

The modelling of a white dwarf's interior would be particularly interesting, as the body is held up against gravitational collapse by electron degeneracy pressure, rather than the thermal and radiation pressure of a main sequence star. Also note that its polytropic index is the exact boundary at which the polytropic model switches from predicting a fully convective to a fully radiative interior (see subsection 4.3). The reason for which white dwarf stars were not considered in this study was largely due to the absence of an easily-accessible standard to which the results could be compared. While theoretically the SSM could be used to evolve the Sun beyond the point at which it becomes a white dwarf, the software required to do this (*e.g.* Modules for Experiments in Stellar Astrophysics — MESA) is complex enough to fall outside the scope of this project.

In relation to the complex solutions of the Lane-Emden equation (section 5), only the domains of the exact solutions were extended to the complex plane. If time afforded it, a more comprehensive analysis in which the computation of solutions with complex domains and polytropic indices $n \notin \{0, 1, 5\}$ would have been attempted. This more liberal type of analysis would also allow for the

alteration of the initial conditions of the solution, whereas this study focused solely on the complex solutions of the Lane-Emden equation which are both physical *and* exact. Furthermore, the nature of the movable poles in the solution corresponding to $n = 5$ (Figure 12) would be elucidated by this analysis, as it is the initial conditions which determines the position of these singularities — and therefore the radius of convergence of a power series centred at any given point.

7 Conclusion

When compared to the standard solar model, the polytrope model was found to accurately model many parameters of the solar interior. This is particularly true when the relative complexities of the polytrope model (which is analytical) and SSM (which is numerical) are contrasted.

The behaviour of the solutions when they are free to become complex, or when their domains can span the complex plane, was also investigated. It was found that for solutions $\theta : \mathbb{R} \rightarrow \mathbb{C}$ with a fractional value of n that the imaginary components depend highly on the polytropic index, being negligible (at low ξ) for a large range of n . Those solutions with $n \geq 5$ never gained any imaginary component, as $\theta \geq 0 \forall \xi \in \mathbb{R}^+$ for these types of functions. When the domain of θ was expanded to the complex numbers, the solution corresponding to $n = 5$ showed movable poles which were absent from the other exact solutions.

Ultimately, modelling stellar objects as polytropes proved to be satisfactorily accurate when compared to the complexity of doing so, and the Lane-Emden equation proved fruitful in terms of its potential for interesting analysis.

References

1. Knight, R. All about polytropic processes. *The Physics Teacher* **60**, 422–424.
<https://www.physics.smu.edu/scalise/P3374fa24/notes/polytropes.pdf> (2022).
2. Emden, R. Anwendungen Der Mechanischen Wärmetheorie Auf Kosmologische Und Meteorologische Probleme. <https://archive.org/details/gaskugelnanwend00emdegoog/page/n5/mode/2up?ref=ol&q=n%3D5> (1907).
3. Hunter, C. Series solutions for polytropes and the isothermal sphere. *Monthly Notices of the Royal Astronomical Society*. <https://doi.org/10.1046/j.1365-8711.2001.04914.x> (2001).
4. Dhillon, V. Solving the Lane-Emden Equation.
https://vikdhillon.staff.shef.ac.uk/teaching/phy213/phy213_le.html#table2 (2012).
5. Buchdahl, H. A. *Remark on the Polytrope of Index 5* 1978.
<https://connectsci.au/ph/article/31/1/115/198505/Remark-on-the-Polytrope-of-Index-5>.
6. Petersen, J. O. *The Eddington-Strömgren-Modelling of constitution and Evolution of Stars from the 1920s and 1930s* 2019. <https://otzenpetersen.dk/ABC/C9.pdf>.
7. Bahcall, J. N., M, S. A. & Basu, S. New solar opacities, abundances, helioseismology, and neutrino fluxes. *The Astrophysical Journal Letters* **621**, L85–L88.
<https://arxiv.org/pdf/astro-ph/0412440> (2005).
8. Bahcall, J. Standard Solar Model Data.
<https://www.sns.ias.edu/~jnb/SNdata/solarmodels.html>.
9. Kennewell, J. & McDonald, A. The Source of Solar Energy. *Australian Space Weather Forecasting Centre*. <https://www.sws.bom.gov.au/Educational/2/1/11>.
10. Facility, A. T. N. Main Sequence Stars.
<https://www.atnf.csiro.au/resources/education/senior-astrophysics/stellarevolution/mainsequence/> (2026).
11. Power Series Expansion for Sinc Function.
https://proofwiki.org/wiki/Power_Series_Expansion_for_Sinc_Function.
12. Lane, H. J. On the theoretical temperature of the Sun, under the hypothesis of a gaseous mass maintaining its volume by its internal heat, and depending on the laws of gases as known to terrestrial experiment. *American Journal of Science* **s2-50**, 57–74 (July 1870).
13. Kycia, R. A. & Filipuk, G. *On the Singularities of the Emden–Fowler Type Equations in Current Trends in Analysis and Its Applications* (eds Mityushev, V. V. & Ruzhansky, M. V.) (Springer International Publishing, Cham, 2015), 93–99. ISBN: 978-3-319-12577-0.
14. Sagert, I., Hempel, M., Greiner, C. & Schaffner-Bielich, J. Compact stars for undergraduates. *European Journal of Physics* **27**, 577–610. ISSN: 1361-6404.
<http://dx.doi.org/10.1088/0143-0807/27/3/012> (Apr. 2006).

Acknowledgements

We would like to extend our thanks to Professor Peter Coles for his guidance concerning the computational methods used in this project, as well as to Dale Lawlor and Saoirse Ward for their help in the lab — particularly for their aid in finding quality references and research materials. Finally, to Professor Emma Whelan, who provided crucial advice in relation to the modelling of stellar parameters and the standard solar model.

Appendix A: Glossary of Symbols

α	The constant of proportionality which links the radial coordinate r with the dimensionless measure of length ξ .
β	The power to which the volume is raised in the polytropic equation of state (Equation 1), is commonly substituted for $1 + \frac{1}{n}$.
γ	The adiabatic index of a gas.
C	The constant of the polytropic equation of state (Equation 1). It has dimensions of $M \cdot T^{-2} \cdot L^{3\beta-1}$.
ϵ	The specific power of stellar material due to nuclear fusion.
θ	A dimensionless measure of density, the proportionality relation between θ and ρ is given by Equation 5. The solutions of the Lane-Emden equation are a set of θ -functions.
k_B	Boltzmann's constant.
μ	The average atomic weight of a gas, usually in the context of stellar material.
M_r	The mass enclosed within a spherical surface of radius r , with its centre coincident with the barycentre of some relevant star.
$M_\odot$	The mass of the Sun, used as the standard unit of mass unless otherwise stated.
m_H	The mass of a hydrogen atom.
n	The polytropic index of a gas which obeys the polytropic equation of state (Equation 2). It is involved in the often-used substitution $\beta \rightarrow 1 + \frac{1}{n}$.
π	The ratio between the circumference and diameter lengths of a circle.
P	Pressure, usually in the context of the stellar interior.
$P_{c,\odot}$	The pressure in the core of the Sun, used as the standard unit of pressure unless otherwise stated.
ρ	Density, usually in the context of the stellar interior.
r	The radial component of a point in a spherical coordinate system.
R	The radius of some relevant star.
$R_\odot$	The radius of the Sun, used as the standard unit of length unless otherwise stated.
T	Temperature, usually in the context of the stellar interior.
$T_{c,\odot}$	The temperature in the core of the Sun, used as the standard unit of temperature unless otherwise stated.
ξ	A dimensionless measure of length, the proportionality relation between ξ and ρ is given by Equation 4.
ξ_R	The first root of the solution to the Lane-Emden equation for a given polytropic index. Some values are displayed in Table 1.
ξ_R^*	The first root of the exact solution to the Lane-Emden for a given polytropic index, should an analytical solution exist. Some values are displayed in Table 1.
$V_\odot$	The volume of the Sun, used as the standard unit of volume unless otherwise stated.

Appendix B: Derivations

B1: The Lane-Emden Equation — Equation 3

Begin with two fundamental equations of stellar structure. Equation 22 is the mass continuity equation and Equation 23 is the equation for hydrostatic equilibrium in a self-gravitating sphere.

$$\frac{dM_r}{dr} = 4\pi r^2 \rho \quad (22)$$

$$\frac{dP}{dr} = \frac{-GM_r \rho}{r^2} \quad (23)$$

From Equation 23

$$\frac{r^2}{\rho} \frac{dP}{dr} = -GM_r \quad (24)$$

$$\Rightarrow \frac{d}{dr} \left(\frac{r^2}{\rho} \frac{dP}{dr} \right) = -G \frac{dM_r}{dr} \quad (25)$$

Filling Equation 22 into this equation

$$\frac{d}{dr} \left(\frac{r^2}{\rho} \frac{dP}{dr} \right) = -4\pi G r^2 \rho \quad (26)$$

Assume a polytrope equation of state, *i.e.* $P = K\rho^{1+\frac{1}{n}}$ where K is a constant and n is the polytropic index.

$$\frac{d}{dr} \left(\frac{r^2}{\rho} K \frac{d}{dr} \rho^{1+\frac{1}{n}} \right) = -4\pi G r^2 \rho \quad (27)$$

Now define a dimensionless variable θ such that $\rho(r) = \rho_c \theta^n$, where ρ_c is the density in the core of the star.

$$\frac{d}{dr} \left(\frac{r^2 \rho_c^{1+\frac{1}{n}}}{\rho_c \theta^n} K \frac{d}{dr} \theta^{n+1} \right) = -4\pi G r^2 \rho_c \theta^n \quad (28)$$

$$\frac{K(n+1)\rho_c^{\frac{1}{n}-1}}{4\pi G} \frac{1}{r^2} \frac{d}{dr} \left(r^2 \frac{d\theta}{dr} \right) = -\theta^n \quad (29)$$

Then introduce another dimensionless variable ξ such that $r = \alpha \xi$, where $\alpha = \sqrt{\frac{K(n+1)\rho_c^{\frac{1}{n}-1}}{4\pi G}}$.

$$\alpha^2 \frac{1}{\alpha^2 \xi^2} \frac{d}{d(\alpha \xi)} \left(\alpha^2 \xi^2 \frac{d\theta}{d(\alpha \xi)} \right) = -\theta^n \quad (30)$$

$$\frac{1}{\xi^2} \frac{d}{d\xi} \left(\xi^2 \frac{d\theta}{d\xi} \right) = -\theta^n \quad \blacksquare \quad (31)$$

B2: Analytic solutions to the Lane-Emden equation

For $n = 0$, Equation 3 reduces to

$$\frac{1}{\xi^2} \frac{d}{d\xi} \left(\xi^2 \frac{d\theta}{d\xi} \right) + 1 = 0 \quad (32)$$

rearranging gives

$$\frac{d}{d\xi} \left(\xi^2 \frac{d\theta}{d\xi} \right) = -\xi^2 \quad (33)$$

which can be integrated to get

$$\xi^2 \frac{d\theta}{d\xi} = -\frac{\xi^3}{3} + C_0 \quad (34)$$

which simplifies to the linear ODE

$$\frac{d\theta}{d\xi} = -\frac{\xi}{3} + C_0 \frac{1}{\xi^2} \quad (35)$$

the initial condition of $\frac{d\theta}{d\xi}(\xi = 0) = 0$ implies that $C_0 = 0$

then, a final integration gives the solution

$$\theta = -\frac{\xi^2}{6} + C_1 \quad (36)$$

The second initial condition gives $\theta(\xi = 0) = 1 \implies C_1 = 1$, and so the final solution becomes

$$\theta(\xi) = 1 - \frac{\xi^2}{6} \quad (37)$$

Next, for $n = 1$, Equation 3 reduces to

$$\frac{1}{\xi^2} \frac{d}{d\xi} \left(\xi^2 \frac{d\theta}{d\xi} \right) + \theta = 0 \quad (38)$$

which, in its standard form, can be written as

$$\frac{d^2\theta}{d\xi^2} + \frac{2}{\xi} \frac{d\theta}{d\xi} + \theta = 0 \quad (39)$$

which is a 2^{nd} order homogeneous differential equation, with a regular singular point at $\xi = 0$. As such, assume a solution of the form $\theta = \sum_{n=0}^{\infty} a_n \xi^{n+r}$, for some $r \in \mathbb{R}$

The differential equation then becomes

$$\sum_{n=0}^{\infty} a_n(n+r)(n+r-1)\xi^{n+r-2} + \sum_{n=0}^{\infty} 2a_n(n+r)\xi^{n+r-2} + \sum_{n=0}^{\infty} a_n\xi^{n+r} = 0 \quad (40)$$

which simplifies to

$$\begin{aligned} a_0[r(r-1) + 2r]\xi^{r-2} + a_1[(r+1)r + 2(r+1)]\xi^{r-1} \\ + \sum_{m=0}^{\infty} [a_{m+2}(m+r+2)(m+r+1) + 2a_{m+2}(m+r+2) + a_m]\xi^{m+r} = 0 \end{aligned} \quad (41)$$

which implies that

$$a_0r(r+1) = 0 \quad (42)$$

$$a_1(r+1)(r+2) = 0 \quad (43)$$

$$a_{m+2} = -\frac{a_m}{(m+r+2)(m+r+3)} \quad (44)$$

As $a_0 = a_1 = 0$ does not constitute a solution, the indicial roots are given by $r = 0$ and $r = -2$.

For $r = -2$, $a_0 = 0$, and the solution is given as

$$\theta = a_0 \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)!} \xi^{2n} = \frac{a_0}{\xi} \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)!} \xi^{2n+1} = a_0 \frac{\sin(\xi)}{\xi} \quad (45)$$

For $r = 0$, $a_1 = 0$, and the solution is given as

$$\theta = a_1 \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n)!} \xi^{2n+1} = \frac{a_1}{\xi} \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n)!} \xi^{2n} = a_1 \frac{\cos(\xi)}{\xi} \quad (46)$$

So, a family of solutions is given by

$$\theta = C_0 \frac{\sin(\xi)}{\xi} + C_1 \frac{\cos(\xi)}{\xi} \quad (47)$$

Now, the initial condition stipulate that

$$\lim_{\xi \rightarrow 0} (\theta) = \lim_{\xi \rightarrow 0} \left(\frac{C_0 \sin(\xi) + C_1 \cos(\xi)}{\xi} \right) = 1 \quad (48)$$

which is satisfied by the integration constants $C_0 = 1$ and $C_1 = 0$

As such, $\theta(\xi)$ is given by

$$\theta = \frac{\sin(\xi)}{\xi} \quad (49)$$

It is important to note that

$$\begin{aligned} \lim_{\xi \rightarrow 0} \left(\frac{d\theta}{d\xi} \right) &= \lim_{\xi \rightarrow 0} \left(\frac{\cos(\xi)}{\xi} - \frac{\sin(\xi)}{\xi^2} \right) = \lim_{\xi \rightarrow 0} \left(\frac{\xi \cos(\xi) - \sin(\xi)}{\xi^2} \right) \\ &= \lim_{\xi \rightarrow 0} \left(\frac{\cos(\xi) - \xi \sin(\xi) - \cos(\xi)}{2\xi} \right) = \lim_{\xi \rightarrow 0} \left(\frac{\sin(\xi)}{2} \right) = 0 \end{aligned} \quad (50)$$

and so the initial condition of $\frac{d\theta}{d\xi} = 0$ is also satisfied by the chosen constants of integration.

Overall, the solution to the LE-equation becomes

$$\theta = \frac{\sin(\xi)}{\xi} \quad (51)$$

Finally, for $n = 5$, Equation 3 reduces to

$$\frac{1}{\xi^2} \frac{d}{d\xi} \left(\xi^2 \frac{d\theta}{d\xi} \right) + \theta^5 = 0 \quad (52)$$

Now, from (Emden, 1908) [2], $\theta = \frac{1}{\sqrt{1+\frac{\xi^2}{3}}}$ is given as a solution to this differential equation. It is only verified here that this is indeed a solution to the Lane-Emden equation of polytropic index $n = 5$.

$$\theta = \frac{1}{\sqrt{1+\frac{\xi^2}{3}}} = \left(1 + \frac{\xi^2}{3} \right)^{-\frac{1}{2}} \quad (53)$$

$$\Rightarrow \frac{d\theta}{d\xi} = -\frac{\xi}{3 \left(1 + \frac{\xi^2}{3} \right)^{\frac{3}{2}}} \quad (54)$$

$$\begin{aligned} \Rightarrow \frac{d^2\theta}{d\xi^2} &= -\frac{1}{3} \left[\left(1 + \frac{\xi^2}{3} \right)^{-\frac{3}{2}} - \frac{3\xi}{2} \left(1 + \frac{\xi^2}{3} \right)^{-\frac{5}{2}} \left(\frac{2\xi}{3} \right) \right] \\ &= -\frac{1}{3} \left[1 + \frac{\xi^2}{3} - \xi^2 \right] \left(1 + \frac{\xi^2}{3} \right)^{-\frac{5}{2}} = \frac{(2\xi^2 - 3)}{9} \left(1 + \frac{\xi^2}{3} \right)^{-\frac{5}{2}} \end{aligned} \quad (55)$$

Thus,

$$\begin{aligned} \frac{d^2\theta}{d\xi^2} + \frac{2}{\xi} \frac{d\theta}{d\xi} + \theta^5 &= \left[\frac{1}{9} (2\xi^2 - 3) - \frac{2}{3} \left(1 + \frac{\xi^2}{3} \right) + 1 \right] \left(1 + \frac{\xi^2}{3} \right)^{-\frac{5}{2}} \\ &= \frac{1}{9} [2\xi^2 - 2\xi^2 - 3 - 6 + 9] \left(1 + \frac{\xi^2}{3} \right)^{-\frac{5}{2}} = 0 \end{aligned} \quad (56)$$

and so $\theta = \frac{1}{\sqrt{1+\frac{\xi^2}{3}}}$ is indeed a solution to the Lane-Emden equation of polytropic index $n = 5$. ■

B3: Uniform Energy Transportation within Polytropes

The condition for convection within a star is

$$\frac{TdP}{PdT} < \frac{\gamma}{\gamma-1} \quad (57)$$

We can assume that solar plasma is so highly ionised that it acts like a monatomic gas (*i.e.* $\gamma = 5/3$).

$$\therefore \frac{TdP}{PdT} < \frac{5}{2} \quad (58)$$

The equations for T and P for a polytrope have already been derived (Equation 13 and Equation 2 respectively).

$$\frac{\left(\frac{\mu m_H}{k_B} \rho^{\frac{1}{n}}\right) d(K\rho^{1+\frac{1}{n}})}{K\rho^{1+\frac{1}{n}} d\left(\frac{\mu m_H}{k_B} \rho^{\frac{1}{n}}\right)} < \frac{5}{2} \quad (59)$$

$$\left(\frac{\mu m_H}{K k_B}\right) \frac{1}{\rho} \left(\frac{K k_B}{\mu m_H}\right) \frac{d(\rho^{1+\frac{1}{n}})}{d(\rho^{\frac{1}{n}})} < \frac{5}{2} \quad (60)$$

$$\frac{1}{\rho} \frac{d(\rho^{\frac{1}{n}})^{n+1}}{d(\rho^{\frac{1}{n}})} < \frac{5}{2} \quad (61)$$

Make the substitution $\rho^{\frac{1}{n}} \rightarrow x$

$$\frac{1}{x^n} \frac{d}{dx} x^{n+1} < \frac{5}{2} \quad (62)$$

$$\frac{1}{x^n} (n+1) x^n < \frac{5}{2} \quad (63)$$

$$n+1 < \frac{5}{2} \quad (64)$$

$$n < \frac{3}{2} \quad (65)$$

So, any polytrope with $n < 1.5$ will be fully convective, and any star with $n > 1.5$ will be fully radiative. ■

Appendix C: Complex Solutions to the Lane-Emden Equation

As mentioned in subsection 3.3, the complex valued solutions to the Lane-Emden equation is an avenue of research which, though of interest, is outside the scope of this study. The following is a glance at possible noteworthy aspects of this open question.

The complex solutions arise due to the θ^n term in the Lane-Emden differential equation. As θ turns negative, taking it to an arbitrary power results in a potential imaginary contribution to the evaluation of the ODE solution. A selection of complex solutions are shown in Figure 13, over a range of polytropic indices. Several domains of interest have been identified.

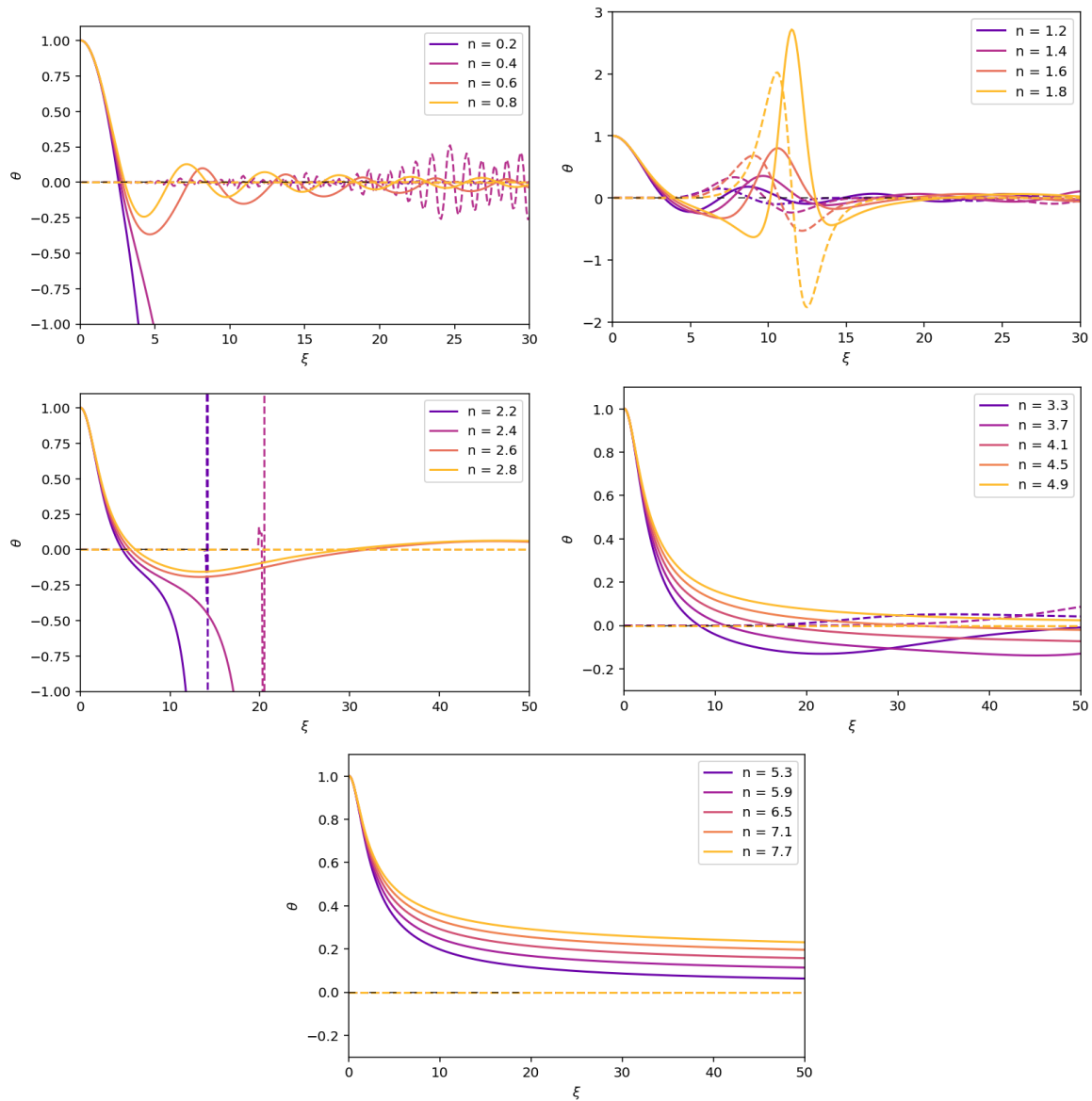


Figure 13: Sample complex solutions to the Lane-Emden equation, for different ranges of polytropic indices - Real (solid) and Imaginary (dashed) parts shown

The following list contains a preliminary analysis of the behaviour of the complex solutions to the Lane-Emden equation, within each range of n .

$$0 < n < \frac{1}{2}$$

- Real part: Steep decline, unknown asymptotic behaviour
- Imaginary part: Oscillatory around $\text{Im}(\theta) = 0$

$$\frac{1}{2} < n < 1$$

- Real part: Oscillatory around $\text{Re}(\theta) = 0$
- Imaginary part: Negligible

$$1 < n < 2$$

- Real part: Brief oscillations, followed by a stabilisation around $\text{Re}(\theta) = 0$
- Imaginary part: Single oscillation, approaching a limit of $\text{Im}(\theta) = 0$

$$2 < n < \frac{5}{2}$$

- Real part: Steep decline, possibly asymptotic
- Imaginary part: Negligible, except singularities present (dependent on initial conditions)

$$\frac{5}{2} < n < 3$$

- Real part: Stabilises around $\text{Re}(\theta) = 0$
- Imaginary part: Negligible

$$3 < n < 5$$

- Real part: Stabilises around $\text{Re}(\theta) = 0$
- Imaginary part: Significant contribution for large ξ

$$n > 5$$

- Real part: Slow decrease, asymptotic to some non-negative $\text{Re}(\theta)$
- Imaginary part: Non-existent

Any further analysis into the non-physical, complex valued solutions to the Lane-Emden equation is left as an open question, for future research.